Reading makes a full man
conference a ready man
and writing an exact man

--- Francis Bacon

Research&
Paper Writing

LWD

- 2025-3

What is a Scientific Paper?

- ***Purpose***:
  Share your own original research work with other scientists or review the research conducted by others,
  promoting the development of modern science.

- ***Characteristics***:
  Need to be highly readable, that is, clear, accurate, and concise.
  Aim to inform, not impress.

- ***Audiences***:
  Referees (help the journal editor decide whether a paper is suitable for publication),
  and journal readers (with different levels of knowledge about the topic).

Why paper?

motivations for writing a science paper.

- Why Write and Publish a Paper?

  two kinds of motivations,

  altruism and self-interest,
- Publication is usually the most straightforward way to make such a contribution.

How start?

The Literature Search

- The goal of the search is to evaluate the state of our communal knowledge on a topic before embarking on a quest of adding to that knowledge.

Because science is about either confirming or refuting existing knowledge or developing new knowledge,

a thorough understanding of the current state of communal knowledge is essential.

Are your results new?

Literature studies **are essential before you start the research program and writing.**

There are hundreds of years of scientific research and millions of papers.

To publish a paper you must be at the *research frontier*.

You must know *where the research frontier is*.

You must continually check the *most recent literature* **to make sure what others are doing.**

Plan and Execute Research with Publication in Mind

document the work in sufficient detail so that

- the reader can follow the reasoning presented and validate the conclusions drawn,
- to defend the work against criticism

Why format? structure and organize?

- Over the last 350 years, scientific journals have evolved a distinctive style, structure, and organization that make it easy for both the writer and the reader to get what they need from the paper: effective communication of scientific ideas
- By following the standard structure and organization of a science research article, the author is constrained in many respects. But these constraints free the author and the reader to focus on the content, which often results in a better paper.

Outline

A good outline for the paper is *a good plan* for the research program.

An outline is a *written plan of the organization* of a paper, including the data on which it rests.

It can be relatively efficient to *go through several (even many) cycles* of an outline before beginning to write text.

The purpose of your paper is **to convey the idea**

- **Here is *a problem***
- **It's an *important problem***
- **It's an *unsolved problem***
- **Here is *my idea***
- **My *idea works* (details, data)**
- **Here's how my idea *compares to* other people's approaches**

Basic parts of a paper

- Title
- Abstract
- Introduction
- Support (theoretical and experimental)
- Results and discussions
- Conclusions
- References-plagiarize, plagiarism, misconduct
- Acknowledgements
- Appendix

- most papers use an “IMRaD” format:

Introduction
Method (experiment, theory, design, model)
Results
and Discussion
Conclusions

- it makes it easier for the writer to organize the content of the paper, and it makes it easier for the reader to opportunistically find the information they seek. T

Introduction

the Introduction section should answer two questions: “

- What?” and “So what?”
- What is the paper about, and why should the reader care?

- scope, novelty, and significance
- Quality is implied and should be demonstrated, not explicitly claimed.

As Swales has described it, the research-article introduction moves through three phases:

- Establish a territory (what is the field of the work, why is this field important, what has already been done?),
- Establish a niche (indicate a gap, raise a question, or challenge prior work in this territory),
- and Occupy that niche (outline the purpose and announce the present research; optionally summarize the results).

three parts of the introduction uses

- topic, problem, solution (for engineering);
- or topic, observation/discovery, explanation (for science).
- Some articles finish the introduction by presenting an outline of the article

Introduction

Style

- Problem X is important
- A, B, and C have been done
- A, B, and C have their weakness (don’t over criticize)
- Our work D
- Features and advantages of D
- Results
- Organization of the paper

As you finish each section, go back to the introduction to revise corresponding contents.

Introduction – function

- Introduce the subject: what is it about
- Limit the research scope: where to start
- Stating the general purpose: where to go
- Explain the writing arrangement: how to go

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Introduction – writing process

- Start with the research background: literature review
- Move towards the existing problem: new / unsolved
- Focus on the present research: objectives, novel ideas, advanced methods, new materials, research scope

Textual development

- Papers of theoretical nature:
  textual description, logical development, reasoning
- Papers of experimental nature:
  experiment process, technology, materials, equipment, conditions, result analyses
- Papers of combined nature:

Some common pitfalls in writing an introduction include

- providing unnecessary background information (telling the reader what they already know or what they do not need to know),
- exaggerating the importance of the work,
- or failing to make clear what research questions this paper is trying to answer.

Method

- describes how the results were generated. It should be sufficiently detailed so that an independent researcher working in the same field could reproduce the results sufficiently to allow validation of the conclusions.

- A good method section should not only describe what was done and how it was done, but it should justify the experimental design as well. Of the many options available, why was this method chosen? Statistical considerations, such as sampling plans and analysis methods used, should be explained. If the raw results are not going to be presented, then a description of the data-reduction procedures is required. Also, consider how a figure or diagram might be used to illustrate or summarize the methods.

- A common shortcoming of method sections in many papers today is the abandonment of the goal of reproducibility.
- Other common pitfalls when writing the Method section: including results in the Method section, including extraneous details (unnecessary to enable reproducibility or judge validity), or treating the method as a chronological history of what happened.

Support (theoretical and experimental)

Purpose: Introduce your work

Theoretical analysis

The methods must be described in such a way that another researcher can *repeat* exactly what is done.

- *Use references to describe standard or published methods*.
- *Note any deviations that you have introduced in this work*.

Support (theoretical and experimental)

Experiments

Style

- *Experimental design*

  Be sure that other researchers can repeat your experiments according to your descriptions.

- *Analysis of the results*
- *Comparisons*

  Put tedious details in Appendix.

Results and Discussion

- Often, these results are presented in tables and/or graphs

- An important goal when presenting results is to clearly designate those results that are new (never before published), while properly citing results that have been previously published

Results

- Choosing only meaningful data: classifying and selecting
- Short and clear: most important but shortest between experiment and discussion
- Avoid redundancy in citing figures:
  - Poor: “It is clearly shown in Table 1 that the information desire grows with the development of industry.”
  - Improved: “The information desire grows with the development of industry (Table 1).”

Discussion

- Analyze data: factual relation, generalization
- Point out doubts: failures, points for attention
- Expound the viewpoint: judgment, evaluation, analysis
- Come to the conclusion: summary of end-result

- The purpose of the Discussion section is to explain the results and show how they help to answer the research questions posed in the introduction.
- This discussion generally passes through the stages of
  1. summarizing the results, discussing whether results are expected or unexpected,
  2. comparing these results to previous work,
  3. interpreting and explaining the results (often by comparison to a theory or model), and hypothesizing about their generality.
- The Discussion section inverts the format of the introduction, moving from the specific (the results generated in this work) to the general (how these results demonstrate a general principle that is more widely applicable).
- Any problems or shortcomings encountered during the course of the work should also be discussed, especially if they might influence how results are to be interpreted.

Some common pitfalls when writing the results and discussion section are

- a lack of organization,
- presenting results that are never discussed, presenting discussion that does not relate to any of the results,
- presenting results and discussion in chronological order rather than logical order,
- ignoring results that do not support the conclusions,
- or drawing conclusions from results without sound logical arguments to back them up

Results and Discussion

Results

This section should simply *state* all results of the experiment.

Do not include information on what you believe should *have happened* or *any analysis* of the results.

The results section can include any *graphs* and *data tables*.

Results

- Organize results neatly if *graphs* can be neatly placed within the text, you may do so.
- If not, *appendices should be used* and referred to in the Results section.
- Be sure to *state* what happened…not what was supposed to happen.
- You should *not draw any conclusions* in the results section.

Discussion

It is used to *explain*, *interpret*, and *analyze* your results. You should explain *errors* or *problems* with the experiment of data and can *compare* your results to previous research and theory.

- Don't merely analyze the results, also discuss the *implications* of those results.
- Pay attention to the *errors* that existed in the experiment.
- Differs from the Results section in that you now *manipulate the data to defend* your theories.

Conclusions

- The Conclusions section provides a brief summary of the results and discussion, but it should be more than a summary.
- After showing how each research question posed in the introduction has been addressed, the implications of the findings should be emphasized, explaining how the work is significant.
- The goal here is to provide the most general claims that can be supported by the evidence.

- This section should be reader-focused, avoiding a list of all the things that “I” or “we” have accomplished.
- The Conclusions section should allow for opportunistic reading. When writing this section, imagine a reader who reads the introduction, skims through the figures, then jumps to the conclusion.

- The conclusion should concisely provide the key message(s) the author wishes to convey.
- It should not repeat the arguments made in the results and discussion, only the final and most general conclusions. While the results and discussion section is often quite long, the conclusion section is generally short.

- The second goal of the conclusion is to provide a future perspective on the work. This could be recommendations to the audience or a roadmap for future work. A small amount of speculation can be appropriate here, so long as it is relevant and clearly labeled as speculative.
- Some common pitfalls when writing the conclusion are
- repeating the abstract, repeating background information from the introduction,
- introducing new evidence or new arguments not found in the results and discussion,
- repeating the arguments made in the results and discussion,
- or failing to address all of the research questions set out in the introduction.

Abstract and Title

"That writer does the most who gives his reader the most information and takes from him the least time."

-- Charles Caleb Colton

Nothing works better than a well-written title and abstract to make sure that the wrong reader does not waste time on the wrong paper, and that the right reader does not mistakenly skip over the right paper.

- The purpose of a title and abstract is often described as "selling" the paper:
- The true purpose of the title and abstract is to get the right people to read your paper
- Thus, the purpose of the title and abstract is matchmaking:

  matching a paper with the right readers, i.e., those who want and need the information contained in the paper.

Abstract function

- A condensed version of a research paper that highlights the major points covered and provides readers with a brief preview of the content and scope of the writing
- For conf organizers to review before accepting the full paper
- For conf participants to determine which session to attend
- Expanding circulation especially electronically

Types of abstract

- Descriptive:

  information, purpose, methods, scope of paper, introduce the subject

- Informative:

  information, purpose, methods, scope of paper, results / findings, conclusions, recommendations

Features of effective abstract

- Uses one or more well developed paragraphs: unified, coherent, concise, and self-contained
- Uses a 3-part structure
  - Introduction / topic sentence
  - Body / supporting sentences
  - Conclusion / concluding sentence
- Follow chronology of the thesis
- Adds no new information
- Uses passive voice

Order of the elements

- 1. Purpose and method of the study
- 2. Results
- 3. Conclusions and recommendations if any

Verb tense

• Background information	• Present
• Major activities	• Past or present perfect
• Methodology	• Past
• Results	• Past
• Conclusion	• Present

Common problems

- Too detailed
- Important information left out
- Too long
- Too short

Tips

- Select important information from the headings and subheadings, table of contents, introduction and conclusion
- Avoid direct copy of sentences
- Avoid reference to other literatures
- Avoid “I” or “we”
- Use non-evaluating language to report rather than comment
- Remember the word limit

The abstract should be a concise, stand-alone summary of the paper that covers the following topics:

- Background/motivation/context,
- Aim/objective(s)/problem statement,
- Approach/method(s)/procedure(s)/materials,
- Results, and
- Conclusion(s)/implications.

Structured Abstracts

- **Background:** What issues led to this work? What is the environment that makes this work interesting or important?
- **Aim:** What did you plan to achieve in this work? What gap is being filled?
- **Approach:** How did you set about achieving your aims (e.g., experimental method, simulation approach, theoretical approach, combinations of these, etc.)? What did you actually do?
- **Results:** What were the main results of the study (including numbers, if appropriate)?
- **Conclusions:** What were your main conclusions? Why are the results important? Where will they lead?

EXAMPLES:

- Background: Photoresist development rate can be defined microscopically (the development rate at a point) or macroscopically (the propagation rate of an average resist height). In the presence of stochastic noise, these two rates will be different.
- Aim: In order to properly calibrate lithography simulators, the difference between these two definitions of development rate should be quantified.
- Approach: Using theoretical derivations and a stochastic (Monte Carlo) resist simulator, the propagation rate of a resist surface is characterized in the presence of stochastic variation in the resist deprotection concentration and a nonlinear development rate response.
- Results: The resulting propagation rate can be more than an order of magnitude higher than for the case of no stochastic noise. Correlation in the development rate creates an effective surface inhibition over a depth into the resist of several correlation lengths.
- Conclusions: The differences between microscopic and macroscopic dissolution rate can have an important effect on how development rate models should be calibrated, depending on their use in continuum or stochastic lithography simulators.

- Photoresist development rate can be defined microscopically (the development rate at a point) or macroscopically (the propagation rate of an average resist height). In the presence of stochastic noise, these two rates will be different. In order to properly calibrate lithography simulators, the difference between these two definitions of development rate should be quantified. Using theoretical derivations and a stochastic (Monte Carlo) resist simulator, the propagation rate of a resist surface is characterized in the presence of stochastic variation in the resist deprotection concentration and a nonlinear development rate response. The resulting propagation rate can be more than an order of magnitude higher than for the case of no stochastic noise. Correlation in the development rate creates an effective surface inhibition over a depth into the resist of several correlation lengths. These results show that the differences between microscopic and macroscopic dissolution rate can have an important effect on how development rate models should be calibrated, depending on their use in continuum or stochastic lithography simulators.

Introduction and Background

Material and Methods

Results

Conclusions and Recommendations

ABSTRACT

Introduction: India currently hosts 49% of the world's diabetes burden, with an estimated 72 million cases in 2019. The complex cascade of events which leads to cellular malfunction in response to high levels of glucose culminates upon formation of advanced glycation end products (AGEs). Accumulation of AGEs in vivo can induce insulin resistance and impair β-cells of islets of Langerhans, resulting in impaired glucose regulation and development of diabetes. Therefore, AGE is an early risk factor for diabetes. Skin autofluorescence (SAF) spectroscopic screening is the emission of light in the UV-visible, near-IR spectral range when biological substrates are excited with light at suitable wavelength.

Materials and methods: In this study, we estimated the subcutaneous AGE level in 197 nondiabetic healthy subjects and then stratified the population into risk groups for deveploping diabetes based on the result.

Results: In our study, a majority of the healthy population (82% subjects) were in the moderate group, followed by 7% and 3% in the mild and severe groups, respectively, Only 8% of the population screened was in the normal group. In subjects older than 50 years, 95.7% had AGE levels in moderate risk.

Conclusion: Autofluoroscence is an easy tool for early diagnosis; however, larger studies are required.

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- Additionally, the structured method of abstract writing also helps to avoid useless but all-too-common phrases like “in this paper” and “we report” or “will be discussed.” The abstract should talk about the work, not about the paper; phrases like “is discussed” turn your abstract into a table of contents rather than a summary of the paper. Do not use the first person (“I” or “we” or “the author”). Also, there is rarely a need to use phrases like “new” or “novel” in the abstract because only the novel results should be mentioned.

CHECK YOUR ABSTRACT

- There is a three-part test that should be applied to your abstract when you are finished:
- 1. Is all of the information in the abstract consistent with what is written in the body of the paper?
- 2. Can all of the information found in the abstract also be found in the body of the paper?
- 3. Is the important information of the paper found in the abstract? Are any key words from the paper missing from the abstract?

- The abstract must be self-contained, and it should generally not contain citations to other papers. If a citation is required (for example, if the paper is a response to a previous publication), then the full citation must be embedded in the abstract. Avoid abbreviations and acronyms in the abstract, but if you insist on including one, spell it out the first time it is used. Do not refer to figures or tables from the body of the paper, or use words or descriptions that will only make sense after the full paper has been read
- keep the audience in mind
- keep the audience in mind

Titles

- working title
- write the title last
- In general, the title should reflect the aim and approach of the work. Depending on the audience (and the specificity of the journal), some of the background may have to be included. Rarely are results and conclusions even hinted at in the title.

Title

- Function of the title:

  generalize the text, attract the reader, facilitate the retrieval
- Writing requirements of the title:
  - Be brief and concise (less than 15 words)
  - Be specific
  - Avoid questions
  - Be parallel

Sample titles

- Poor: *On the Addition to the Method of Microscopic Research by a New Way of Producing Color: – Contrast Between an Object and its Background or Between Definite Parts of the Object Itself*
- Improved: *A New Way of Producing Color – Contrast in Microscopic Examination*

Sample titles

- Poor: *Computer Retrieval*
- Improved: *Computer Retrieval on Conference Information*
- Poor: *The Design and Preparing of User Documents*
- Improved: *The Design and Preparation of User Documents*

- Unlike the worlds of newspaper reporting and marketing press releases, the title of a scientific paper should describe the aim of the work, not the results. Thus, a good title might be

Impact of temperature and pressure on the compositional uniformity of sputter-deposited aluminum alloys

- The following news-style title, on the other hand, is not appropriate:

Optimizing temperature and pressure improves sputter-deposited aluminum alloy films

- Note that the good title is essentially a statement **of the aim** of the work. Often, it is important to mention **the approach** used as well, though an experimental approach is generally assumed if it is not mentioned. If the study had been based on simulation (or some other approach), however, this would generally be included in the title:

Impact of temperature and pressure on the simulated compositional uniformity of sputter-deposited aluminum alloys

- A conflicting goal of the title is to be as short as possible (in 2011, titles in the Journal of Micro/Nanolithography, MEMS, and MOEMS ranged **from 4 to 21 words in length, with an average length of 11.5**). Specificity can often be improved through the use of more words, but a title that is too lengthy may not be read. Finding the best compromise between **descriptiveness and brevity** is where the art of authorship comes into play. Going back to our example, here is a title that sacrifices too much specificity to obtain brevity:

Impact of process parameters on the uniformity of aluminum alloys

- A good test for your title is to answer these questions: Does the title of your manuscript, seen in isolation, give a full yet concise and specific indication of the work reported? Would someone interested in the exact topic of your paper, reading this title, be inclined to read the abstract? Avoid being overly clever with the title—a pun or a play on words may be great fun, but it is unlikely to help your article be found by a search engine (and can be easily misunderstood by an international audience). Titles should also be free of jargon unlikely to be understood by those not intimately familiar with the topic, and they should not contain acronyms. The overall goal should be a title that is clear and informative.

Author / Affiliation

- Function:

  bear responsibility, facilitate retrieval & correspondence, heighten celebrity

- Eg:

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Keywords

- Keywords (also called “subject terms”).
- How you use search engine to locate relevant papers?
- The most important keywords should be found in the title, and in the abstract several times.

Keywords

- Function: retrieve; highlight the gist of paper
- Requirements:
  - Use nouns, not verbs
  - Number of keywords: 4 to 6
  - Chosen from title and abstract
  - No capitalization, use comma or larger space, no full stop
  - Use standard abbreviation: ISO norm

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Acknowledgment

- To extend thanks an indebtedness to people who have helped during the research.
- Supervisors, colleagues, peer students, funding or sponsoring associations, etc.
- Requirements:
  - Be substantial and specific but brief
  - Avoid understatement or overstatement
  - Obtain permission

References

- 统一格式。
- 使用你打算投稿的机构规定的格式。
- 文献的多少要恰当：不要漏掉重要和必要的文献，又不要罗列多余的、特别是你自己的但又毫不相关的文章。
- 文献中的每一篇文章都要引用到，否则就不要罗列。

References / Bibliography

- Functions:
  - Show respect to previous works
  - Present scientific bases for the research
  - Avoid legal problem
- Sample :
  - In text: (Smith, 1983: 123)
  - End of paper: Smith, L., 1983, *Reading in English as an International Language*, Oxford: Pergamon.

Appendix

附录通常可以放一些比较长的引理和定理的证明。

这样会方便读者顺利地阅读文章的全文，而只在有必要的时候才去检查这些证明。

Appendices

- For any materials which are important but do not fit naturally in the text.
- Materials included
  - Questionnaires used to obtain data
  - Sheets used in observation
  - Photocopied documents such as textbooks, exam papers and syllabuses
  - Other materials that are long and awkward if put in the text

Appendices – requirements

- All materials in an appendix must have been referred to in the text
- Each document should be included in a separate appendix which should be numbered and titled and begin on a new page

Misconceptions 1:

The more, the better

Don’t provide too many “new” ideas in a paper. Try your best to focus on your core work.

Misconceptions 2:

The more complex, the better

Try to make your paper easy to be understood.

Remember: You are showing your thoughts instead of confusing the readers.

Mathematical language is to help you clearly express your thoughts, not to show your depth.

Misconceptions 3:

The more selling, the better

Don’t exaggerate too much on your work.

The reviews/readers will judge it.

Misconceptions 4:

The more authoritative, the better

Don’t refer too many own works while ignoring others.

Don’t behave as a big shot if you are not.

Tricks

· Read more, write more

. Use good papers as samples

to simulate but not to copy

· Use simple sentences

· Not to show your elegance

unless you are very famous

· Try to reduce syntax errors

Good works won’t be rejected for syntax, but at least the paper should be readable.

Tricks

. Make your abbreviation easy to be pronounced

· Typeset according to the requirements of the target

· Revise, revise, and revise

· Improve from comments

Thank You !